Methodology for Manufacturing Improvement

Jiri Tupa[1], Ales Hamacek[1], Jan Reboun[1], Vlastimil Skocil[1], Frantisek Steiner[1]

[1] University of West Bohemia in Pilsen, Department of Technologies and Measurement
Univerzitni 26, CZ 306 14 Pilsen, Czech Republic
tupa@ket.zcu.cz

***Abstract**: The paper deals with the new CQT methodology approach to the improvement of the process control in printed circuit board (PCB) manufacturing. The CQT methodology is based on the application of the selected methods and tools. The applied methods cover two key areas. The first area contains methods for description and modelling of processes. The second area contains methods for optimizing of processes. The used tools are focused on performance measurement and optimizing of process (optimizing of cost, process time and quality). This methodology supports process control towards quality improvement, cost and processes time reduction. The paper shows main features of the CQT methodology on the practical example of the PCB manufacturing.*

1. INTRODUCTION

Process control or process management (PM) is a way of production and non-production activities management in manufacturing systems. The process control could effectively be applied in technological process control. A major advantage of this approach, when compared to other approaches, is in the management and control of the interactions between these processes and the interfaces between the functional hierarchies of the organization. This approach is a powerful way of organizing and managing how work activities create value for the customer and other interested parties. The process control introduces horizontal management, crossing the barriers between different functional units and unifying their focus to the main goals of the organization. It also improves management of process interfaces.

These processes control could be defined like processes that transform suitable inputs into needed outputs under specified conditions using available sources. This approach of technological process control can be used in the area of electronic equipments and components manufacturing too. The purpose of the process approach is to enhance an organization's effectiveness and efficiency in achieving its defined objectives. The standard ISO 9001:2000 for quality management system includes this process approach for quality improvement of all enterprises processes too.

1.1. Definition of problems

The present state and basic problems of manufacturing improvement and process control were indicated on the basis of contemporary literature [e.g. 1-4] and practice application. The core problem visualization of the manufacturing process control could be effectively done in the form of the Current Reality Tree (CRT). It is one of the powerful tools of the Theory of Constraint (TOC) methodology for process optimization.

This diagram shows causality of relevant undesirable effects of the analyzed situation. The practical example shows the figure (Fig 1). To solve the main problem – the lack of continues process improvement – the suitable methods and tools have been selected. The description of their exploited features and the common concept of the CQT methodology is main part of this article.

The main contribution of undertaken analysis and the selection process of suitable methods and tools is proposal of the new methodology where the methods and tools are applied in defined steps. The methodology is named CQT according to main goal criteria, eg. C – cost, Q – quality and T – time.

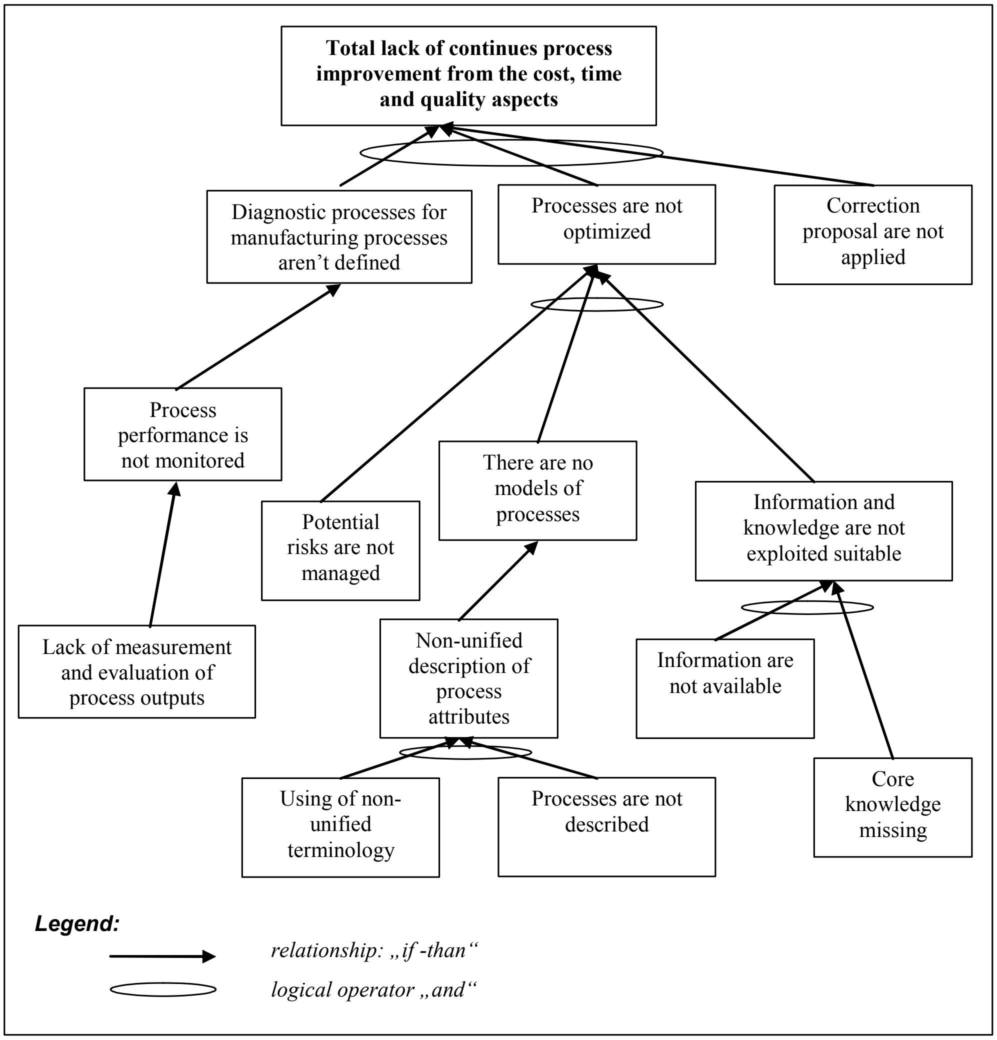


Fig. 1. Current problem causality.

2.1. CQT Methodology

CQT methodology can be used for manufacturing improvement as well. The CQT methodology was developed as one result of our research. This methodology combines known principles and methods the business process management.

The CQT (Cost-Quality-Time) methodology focus to quality improvement, processes time and cost reduction as process goal of processes improvement. The CQT methodology using mathematical methods of graph theory and discrete optimizing for planning and decision tasks, TOC method for optimizing of process time and cost and identification of key manufacturing processes, process analysis and fishbone diagram for quality improvement, risk analysis for process risk management, BSC method and ITIL library as method for diagnostic processes establishment and performance measurement. Table 2 describes the steps of methodology application. The methodology contains all phases of P – plan, D – doing, C – control, A - action cycle (Deming cycle) too (Tab. 1).

Process	P	D	C	A
Current state	X			
Process mapping	X			
Process modeling	X	X		
Key process identification	X			
Key process analysis	X			
Cost and Time analysis	X			
Improvement proposal	X			
Information and knowledge control			X	
Risk analysis			X	
Diagnostic of process				X

Tab. 1. CQT methodology and PDCA cycle.

Process	Input	Output	Method and Tools
Current state and problem identification	Strategy and targets	Current Tree Reality, Future Tree Reality, Conflicts Diagrams and key problems identification	Thinking Process - Current Tree Reality, Future Tree Reality, Conflict Diagram (Evaporating Cloud)
Process mapping	Key problem	Process map, terminology list	Forms for process map construction and process attribute identification
Process modelling	Process map, terminology list	Object process model and process attribute description	Methodology for process modelling e.g. ARIS
Identification of key process making constraint system	Object model	The constraint is determined	TOC (Theory of Constraint) method, mathematical tools e.g. graph theory
Detailed key process analysis	Information about key process	The process was analyzed and information was recorded	Brainstorming, Causes and events diagram, detailed model of process.
Cost and time analysis	Attributes description, accounting, data	Cost and process time analysis	Cost and time analyzing methods, controlling, process measurement
Proposal for improvement and optimizing process according to quality requirements	Process, cost and time analysis	The problem was solved	TOC, real experiment or process testing, mathematical optimizing (application graph algorithm)
Information and knowledge management system and establishment	Optimizing processes	Application of methods and tools	ITIL (Information Technology Infrastructure Library) Information management and knowledge method (e.g. BS 7799)
Risk analysis	Process and situation description	Risk evaluating a risk treatment plan	Risk analysis method
Process diagnostic system design	Process goal and strategy	Performance measurement system and diagnostic process establishment	Balanced Scorecard method ITIL

Tab. 2. CQT methodology.

Glossary of terms and definition is part of this methodology. The glossary contains selected and using terms and their definition for example: process, system, process input, output, process performance, process owner. The objective of this glossary makes to easier process control implementation in real manufacturing.

3. APPLICATION FOR PCB MANUFACTURING

We applied and verified the CQT methodology on the real PCB manufacturing. The PCB manufacture is part of our department. The manufacture produces the one-layer and two-layer PCB. Optimizing problem of production run was main goal of CQT methodology implementation.

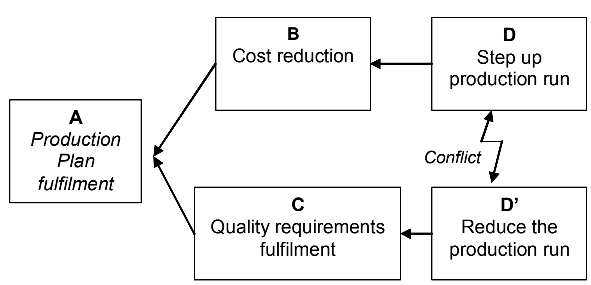


Fig. 2 Conflict diagram of PCB manufacture.

Process identification map record

Processes	Sub processes	Process Type: Main	Supportive	Control	Key	Hard	Soft	Input	Output	Key process measured parametrs
Material Cutting		X				X		Requirements	Material on required dimension	Process time
Drilling PCB	*Drilling*	X			X	X		Material on required dimension	Drilled PCB	Process Time, Failure drilled PCB Number, Quality
	Inspection		X				X	Drilled PCB	Tested PCB	Process Time, Cost
	Diagnostic Planning			X			X	Testing requirements	Diagnostic Plan	Process Time
	Diagnostic Process		X				X	Diagnostic Plan	Diagnosis	Process Time, Cost
Plating Preparing		X			X			Drilled PCB	Prepared PCB	Process Time, Failure PCB number, Quality
Plating PCB		X			X	X		Prepared PCB	Plated PCB	Process Time, Failure PCB number, Quality

PCB Printed Circuit Board

Tab. 3. Process mapping form.

The main optimalizing criteria according to methodology were cost reduction, quality improvement and time reduction. We used the thinking analysis for mapping current situation according to Theory of Constraint (TOC). Results were the Current Tree Reality, Conflict Diagrams (Fig. 2) and Future Tree Reality.

The short demonstration contains the selected parts of CQT methodology, which can be used for manufacturing improvement

3.1. Process mapping

The form represented by Table 3 contains results of process mapping and identification. A process analysis was used for process mapping. Process analysis means identification and description of all processes in the organization, determination of key controllable and measurable parameters and definition of all process attributes. The processes should be dividing on serials, parallels, hard, soft. The hard process means process, where the steps of process can not be exchange. The soft process means process, where the steps of process can be exchange.

3.2. Process modeling

The record of process map should be used for process model construction. We used object modelling by ARIS method (**A**rchitecture **I**nformation **S**ystem). The Figure 3 represents model value – added chain diagram. This model type - specifies the functions in an organization which directly influence the real added value of the organization.

These functions can be linked to one another in the form of a sequence of functions and thus form a value-added chain. The graph model is generated from value – added chain diagram too (Figure 3). The graph model can be used for application mathematical optimization of processes (Figure 4) as well.

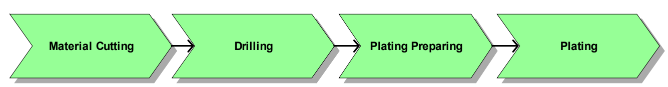
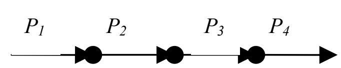


Fig. 3. Process added chain diagram.



Edges: processes, Nodes – states between processes
P_1 Material Cutting, P_2 Drilling, P_3 Plating Preparing, P_4 Plating

Fig. 4. Process graph model.

3.3. Key process identification and analysis

A key process is critical process in manufacturing. The key process was identified by TOC method. The application of TOC used very simple steps:

1. Identify the system's constraint.

2. Decide how to exploit the system's constraint.

3. Subordinate everything else to the above decisions.

4. Elevate the system's constraint. If in any of the previous steps, the constraint has been broken.

The parameter for identification the system's constraint was process time and capacity of process. The process drilling had the worst process time and capacity of all manufacturing processes in this case. So according to CQT methodology the process drilling was analyzed and optimized.

A fishbone diagram was used for technological analysis of drilling process. The optimizing problem was identified from fishbone diagram. It was optimizing the tool life and was determined cycle of drill repair and time to change. The problem has relationship with materials, conditions and methods drilling. Especially the change drill bit geometry and tool life has influence on quality of process. And the tool life is an economic factor which is generally small relative to other cost of production, because machining parameters giving ultra long life tend to reduce productivity. Therefore, it is high productivity that determines the operating parameters, while the operating environment determines the tool life. It is knowledge of the relationship between actual toll life and the operating parameters which permits the establishing of drill change criteria.

3.4. Process model

The eEPC (Extended Event-driven Process Chain) process model ARIS used for modelling of process drilling (Figure 5). Left side of model presents elementary procedures of process (function drive by event). The sub process is linked with important attributes (risk factor, requirement knowledge, data for process, input and output from process). The right side is the example of diagnostic process model for process drilling. For this model was used part of ITIL library. The Incident and Problem management defined by ITIL can be used for diagnostic process control as well.

The ITIL is the documentation of best practice for

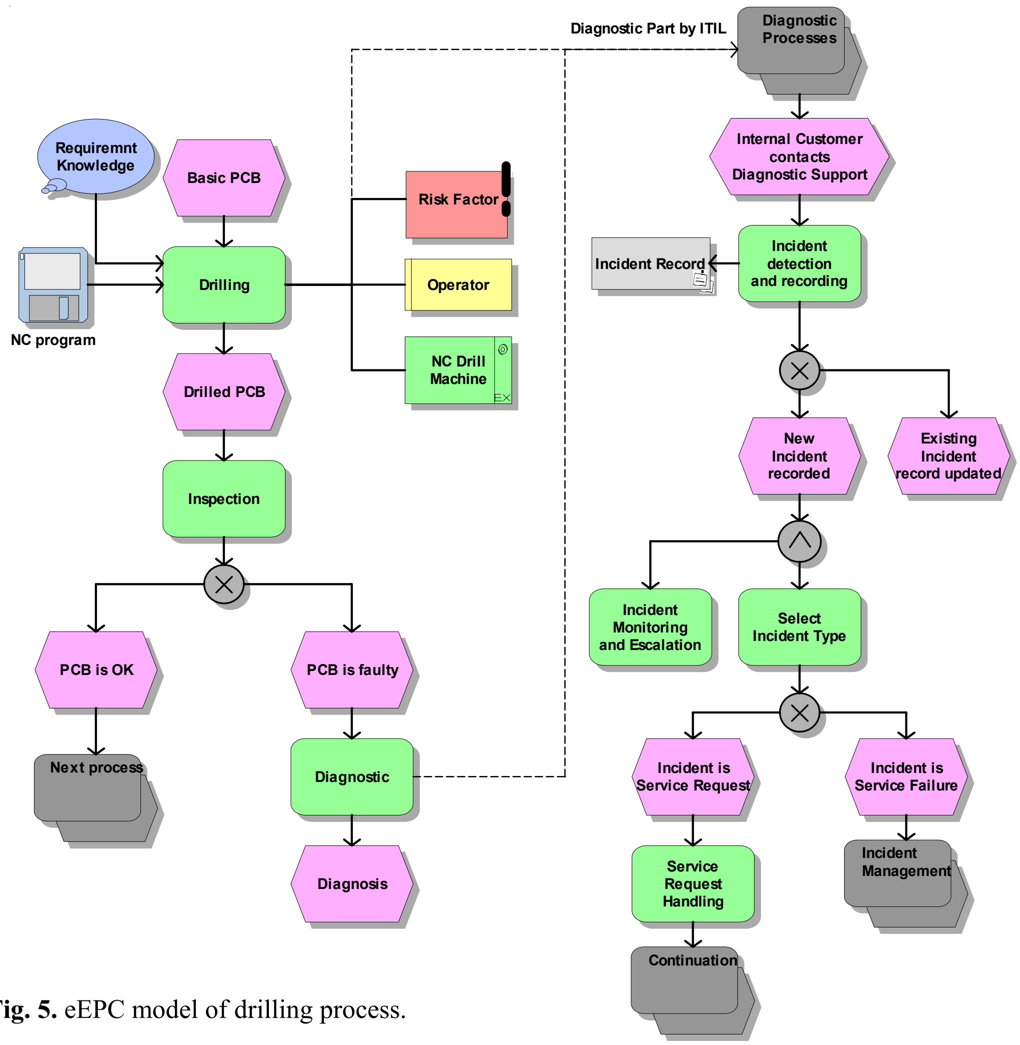


Fig. 5. eEPC model of drilling process.

Information Technology Service Management. The problem and incident management is part of service Support disciplines. The goal of Incident Management is to restore normal service operation as quickly as possible and minimise the adverse impact of Incidents on business operations, thus ensuring that the best possible level of service quality and availability is maintained. Incident Management must be quick as possible. Incident is any event which is not part of the standard operation a service and which causes, or may cause an interruption to, or a reduction in, the quality of the service.

Main goals of Problem management are: to minimise the adverse impact of Incident and Problems on the business; to prevent recurrence of Incidents related to these errors.

3.5. Diagnostic system and performance measurement

The diagnostic system for processes management contains these activities: performance measurement system establishment, diagnostic process design.

Performance measurement system

The performance measurement is important part of PM. Therefore all strategy goals should be determined by measurable metrics or indicators. The strategy goals and metrics should be transferred on all level to all processes (local level) from strategy to global level. One effective of the tools for performance management is Balanced Scorecard (BSC) method. The Balanced Scorecard (BSC) supplements traditional financial measures with three additional perspectives: the customers, the internal business process and the learning and growth perspective.

Diagnostic process establishment

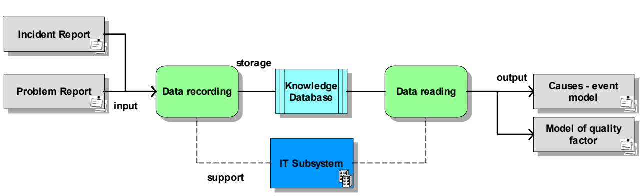


Fig. 6. Process model of knowledge database formation.

The library ITIL can be used for diagnostic process control. The examples of ITIL application was described by previously part. The Figure 6 shows processes a formation of knowledge databases.

The output from knowledge database is model of causes and events e.g. (see Figure 7). Service team use this model as support the incident or problem classification and diagnosis decision-making.

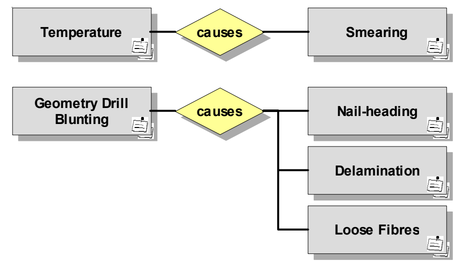


Fig. 7. Model causes and events.

CONCLUSION

This paper presented new CQT methodology for process control of manufacturing improvement. This methodology was applied on example of printed circuit board manufacturing (PCB), especially on process drilling of PCB. The benefits of CQT methodology are: support of business process management systems, quality improvement, cost and process time reduction, focus on diagnostic and performance measurement of processes.

ACKNOWLEDGMENT

This research is supported by the research plan MSM4977751310 “Diagnostic of interactive processes in electrical engineering“.

REFERENCES

[1] Tupa J., Basl J., Steiner F., Skočil V. “Measurement of manufacturing processes”. *Proceedings of 1st International First Manufacturing Engineering Society International Conference*, Cádiz, Spain , pp. 1-8, 2005.

[2] Scheer A.-W., Abolhassan F., Jost, W., Kirchmer M., “Business Process Excellence”, *Springer*, 2002.

[3] http://www.itil.co.uk

[4] Ross W. M.: “Comprehensive Guide to The Design and Manufacture of Printed Board Assemblies”, *Volume 2, Manufacture, Quality Assurance and the Environment*, Port Erin, 1998.

[5] Tupa J., New CQT Technological Process Methodology”, *Doctoral Thesis*, UWB in Pilsen, 2006.